MIL-T-47012 (MI)
19 April 1974
SUPERSEDING
MIS-12802A
25 November 1969

MILITARY SPECIFICATION

TAPE, PRESSURE-SENSITIVE ADHESIVE, COPPER FOIL

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines one type of copper foil electrical tape with an electrically conductive pressure sensitive adhesive.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for porposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated Triple-Wall

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 8030

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Military

MIL-STD-202	Test Methods for Electronic and Electrical Component

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials

ASTM D 1000	Pressure-Sensitive Adhesive Coated Tapes Used for Electrical Insulation, Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Pilot lot sample. Unless otherwise specified (see 6.1), a pilot lot sample of tape shall meet the requirements of this specification.

3.2 Material. The tape shall consist of 0.0014 inches thick (1 ounce) copper foil, coated on the side with an electrically conductive pressure-sensitive adhesive. Unless otherwise specified (see 6.1), the length and width shall be as specified in the contract or order.

3.3 Mechanical properties.

3.3.1 Tensile strength. The tensile strength of the tape shall be not less than 25 pounds per inch.

3.3.2 Adhesion.

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3.3.2.1 After room temperature exposure. Under ambient temperature exposure, the tape shall have an adhesion level of 30 ounces per inch.

3.3.2.2 After high temperature exposure. Tape shall be aged at 130 degrees Celsius (C), plus or minus 3 degrees C for 60 plus or minus 5 hours and shall have an adhesion level of not less than 45 ounces per inch of width, measured at room temperature. Tape shall be applied to substrate before aging.

3.4 Physical properties.

3.4.1 Thickness. The total thickness of the tape shall be not greater than 0.0035, plus or minus 0.0005 inches.

3.4.2 Separator. A liner or separator shall be interposed between adjacent layers or tape within the roll.

3.5 Electrical requirements.

3.5.1 Resistance. Using a contact pressure of 5 pounds per square inch (psi) resistance shall be found by substituting the measured values in Figures 1 and 2 into the formula.

$$R = \frac{V}{I}$$

Where R = Resistance in ohms

V = Voltage in volts

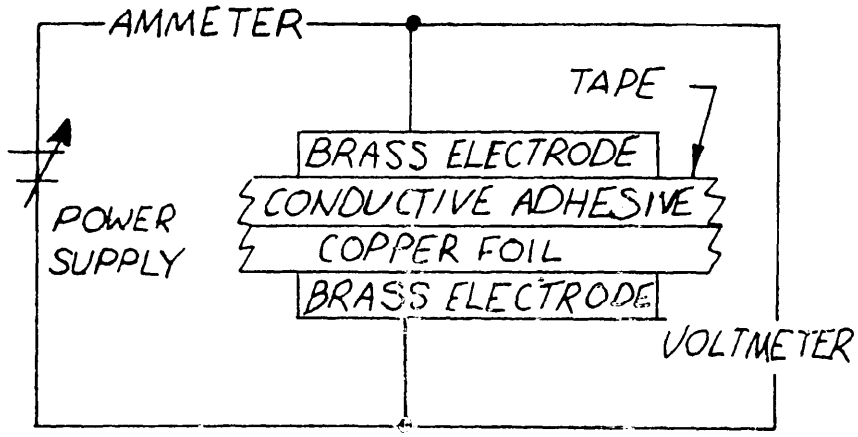
I = Current in amperes

Resistance through the tape shall be .01 ohms per square inch maximum, when tested in accordance with Figure 1, using 1 square inch brass electrodes. Resistance along the tape shall be 0.0065 ohms per inch, maximum when tested in accordance with Figure 2.

3.6 Shelf life. The shelf life of the tape in the packaged condition shall be not less than one year from date of manufacture, when stored at a temperature less than plus 27 degrees C (plus 80.6 degrees F).

3.7 Workmanship. The workmanship of the copper foil tape shall be such as to ensure a high quality product which is uniform and in conformance with the requirements of this specification. The copper foil tape finish shall be smooth, uniform, free from dirt, lumps or bare spots, and shall be free of foreign material or other contaminants that may affect appearance or serviceability.

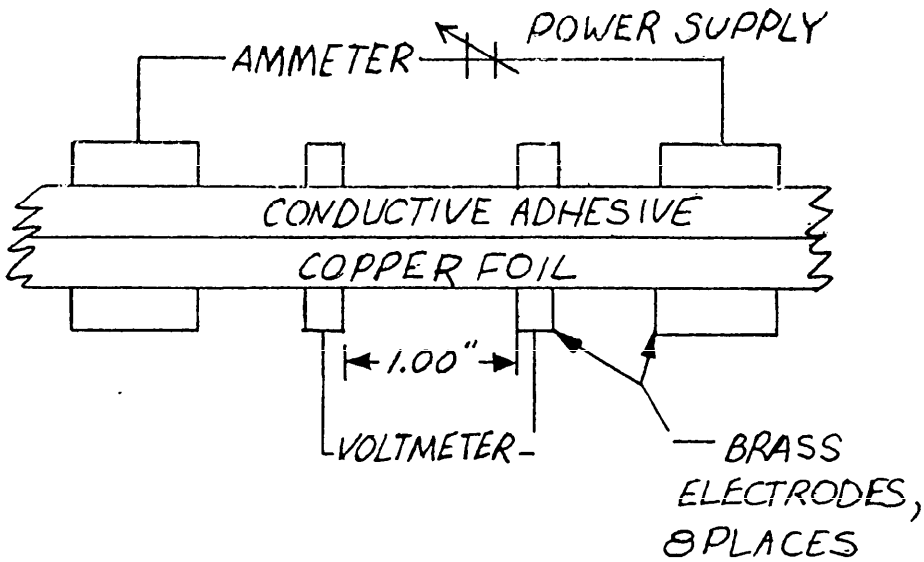
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The power supply is adjusted to deliver 0.1 amperes, and the voltage drop between the electrodes is measured.

Figure 1. Through Tape Resistance Test

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Note 1. Tape Width: 1 inch.

Note 2. The power supply is adjusted to deliver 0.1 amperes. Voltage is measured between two electrodes spaced 1 inch apart.

Figure 2. Along Tape Resistance Test

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Pilot lot. The pilot lot (see 6.1), shall be prepared using the same methods proposed for the preparation of subsequent production lots. When pilot lot specimens do not meet all the requirements of this specification, the entire pilot lot shall be rejected and returned to the supplier. Subsequent lots shall not be considered for acceptance until approval of a pilot lot has been obtained.

4.3 Classification of tests. The examination and testing of the tape shall be classified as follows:

a. Pilot lot tests.

b. Quality conformance tests.

4.3.1 Pilot lot tests. Pilot lot tests shall be conducted only on the pilot lot samples and shall consist of all the examinations and tests specified herein.

4.3.2 Quality conformance tests. Quality conformance tests for acceptance of the tape shall consist of the following examinations and tests:

a. Workmanship.

b. Total thickness of the tape.

c. Thickness of copper foil.

d. Adhesion level without high temperature conditioning.

4.3.2.1 Lot size. Lot size shall consist of all the tape manufactured at one time, from one batch, forming parts of one contract or order, and submitted at the same time and place.

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4.3.2.2 Sampling. Unless otherwise specified (see 6.1), tape representative of each lot shall be selected at random for quality conformance testing. Each roll of tape shall be considered as a unit of product. Failure of sample to meet the quality conformance requirements shall be cause for lot rejection.

4.4 Test methods.

4.4.1 Mechanical properties.

4.4.1.1 Tensile strength. The tensile strength shall be determined in accordance with ASTM D 1000. The tensile strength shall be as specified in 3.3.1.

~~4.4.1.2 Adhesion. Adhesion shall be determined in~~ accordance with ASTM D 1000, using test specimens 10.000 plus or minus 0.126 inch long and 1.000 plus or minus 0.032 inch wide. The adhesion level shall be as specified in 3.3.2.1 and 3.3.2.2.

4.4.2 Physical properties.

4.4.2.1 Total thickness. The thickness of the tape, including foil and adhesive shall be in accordance with ASTM D 1000. The total thickness shall be as specified in 3.4.1.

4.4.2.2 Foil thickness. The foil thickness shall be in accordance with ASTM D 1000. The foil thickness shall be as specified in 3.2.

4.4.3 Electrical test.

4.4.3.1 Resistance. Resistance shall be determined in accordance with MIL-STD-202, Method 303, using test set up per Figures 1 and 2. Resistance shall be as specified in 3.5.1.

4.4.4 Shelf life. The tape shall be tested for conformance with all requirements specified herein after completion of the storage period specified in 3.6. The supplier's certificate of compliance with 3.6 may be accepted during the storage period.

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4.5 Preservation, packaging, packing, and marking. The preservation, packaging, packing and marking shall be examined to determine compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Unless otherwise specified (see 6.1), preservation, packaging, packing, and marking shall be as specified herein.

5.1.1 Preservation and packaging.

5.1.1.1 Level A. Packaging of the tape shall be in accordance with MIL-P-116, Method III. Rolls of tape shall be inserted into weather resistant containers conforming to PPP-B-636.

5.1.1.2 Level B. Same as Level A.

5.1.1.3 Level C. Specified quantities of tape shall be packaged in domestic containers conforming to PPP-B-636.

5.2 Packing.

5.2.1 Level A. A specified quantity of tape packaged as specified shall be packed in a snug-fitting shipping container conforming to PPP-B-640, Class 2, or a weather resistant container conforming to PPP-B-636.

5.2.2 Level B. Same as Level A.

5.2.3 Level C. Tape packaged as specified, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with the rules and regulations of the carrier applicable to the mode of transportation.

5.3 Marking. In addition to any special markings required by the contract or purchase order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

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b. Whether a pilot lot sample is required, and if so, pertinent details (see 3.1).

c. Sampling plan if other than specified (see 4.3.2.2).

d. Requirements for preservation, packing, and packaging (see 5.1 and 5.2).

e. Applicable level(s) of preservation, packaging, and packing (see 5.1.1 and 5.2).

f. Length and width required (see 3.2).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-12802A, dated 25 November 1969.

Custodian:
ARMY-MI

Preparing Activity:
ARMY-MI
Project No. 8030-A046

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